

Real-world data sources

Trace-driven preparation

CA-DTNet core

Digital-twin outputs and validation

pNEUMA traffic trajectories

1-s vehicle trajectories
speed • position • count • flow
road/lane spatial topology

Traffic graph construction

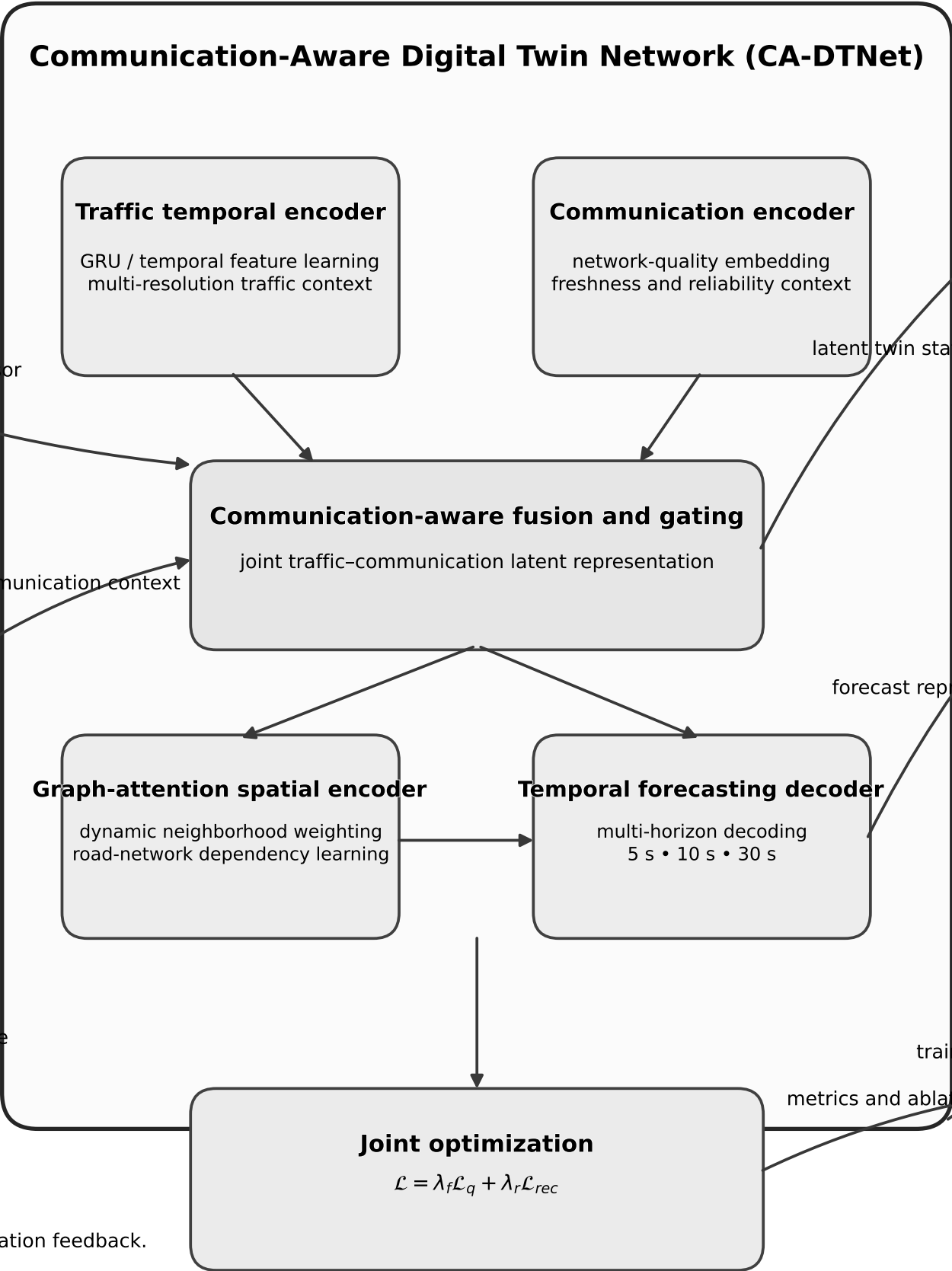
node/road aggregation
adjacency matrix
1-s sliding windows

Vehicular communication traces

CICV5G and SEE-V2X
latency • AoI • packet loss • jitter
channel busy • throughput • RSRP • SINR

Trace synchronization and replay

normalize communication variables
align traces with traffic windows
generate clean and impaired observations



Digital-twin reconstruction head

recover clean traffic state
reconstructed speed/count/flow
state-recovery objective: $\mathcal{L}_{rec}$

Probabilistic forecasting head

median and 10-90% quantiles
speed • vehicle count • flow
quantile objective: $\mathcal{L}_q$

Evaluation and deployment

MAE • RMSE • WAPE • R^2
PICP • PINAW • reconstruction R^2
matched robustness • latency • memory

Solid arrows indicate the forward data path; the lower path represents training, evaluation, and ablation feedback.